

Michael Dilozenzo

Senior Data Engineer

Roseville, CA • michael.dilozenzo.pg@gmail.com
+ 1 (385) 287-0700

PROFESSIONAL SUMMARY

Senior Data Engineer with **10+ years** of experience architecting **real-time, large-scale, and cloud-native data platforms** across **fintech, healthcare**, and regulated industries. Deep expertise in **Spark-based data engineering**, streaming **architectures**, and **modern lakehouse platforms** using **Databricks, Delta Lake, Apache Iceberg**, and **Snowflake**. Proven leader of enterprise data modernization initiatives, unifying batch and streaming systems, implementing metadata-driven ingestion, and enabling **ML and analytics teams** with **reliable, production-grade data pipelines**. Known for strong cross-functional leadership, mentoring engineers, and partnering with compliance, clinical, and product stakeholders to deliver secure, scalable, and audit-ready data ecosystems.

TECHNICAL SKILLS

- **Data Engineering:** Apache Spark (PySpark, Structured Streaming), Kafka, Airflow, dbt, Delta Live Tables
- **Lakehouse & Warehousing:** Databricks, Delta Lake, Apache Iceberg, Snowflake, Redshift, BigQuery
- **Cloud Platforms:** AWS (S3, EMR, Glue, Lambda, IAM, Lake Formation), Azure (Data Factory, Synapse, Blob Storage, Event Hubs), GCP
- **Programming:** Python, SQL, Scala / Java
- **Governance & Quality:** OpenMetadata, Great Expectations, data contracts, lineage, HIPAA, SOC 2, CI-DSS
- **ML & Analytics:** Feature stores, real-time features, Tableau, Power BI, Looker
- **DevOps & Ops:** CI/CD, Terraform, Docker, Kubernetes, Prometheus

PROFESSIONAL EXPERIENCE

Nutanix
Senior Data Engineer

San Jose, CA
May 2022 – Present

- Served as the **Senior Data Engineer** for the design, implementation, and evolution of a **cloud-native, real-time data platform** supporting **regulated fintech and healthcare workloads** across multiple domains.
- Architected a **low-latency streaming architecture** using **Apache Kafka, Spark Structured Streaming**, and **AWS Glue**, enabling **near-real-time ingestion, enrichment, and scoring** of high-volume transactional data.
- Designed and implemented an enterprise **lakehouse architecture** using **Delta Lake on Amazon S3**, providing **ACID-compliant storage**, scalable analytics, and a **unified batch and streaming data layer**.
- Led the migration of **200+ legacy ETL pipelines** to modern orchestration and transformation frameworks using **Apache Airflow** and **dbt**, improving **observability, modularity**, and reducing **SLA violations by 70%+**.
- Built a **metadata-driven ingestion framework** leveraging **Python, YAML**, and **Jinja2**, standardizing ingestion logic, enforcing **schema governance**, and accelerating **source onboarding**.
- Defined and enforced **data contracts, schema evolution policies**, and **versioning strategies** to ensure stability across **real-time and batch pipelines**.
- Implemented scalable ingestion pipelines on **Databricks** using **Auto Loader** and **Delta Live Tables**, supporting **high-throughput, schema-evolving datasets**.
- Integrated **OpenMetadata** for **data lineage, ownership, and discovery**, and **Great Expectations** for automated **data quality validation and profiling**.
- Led **Snowflake performance optimization** using **clustering keys, materialized views**, and **warehouse right-sizing**, reducing **compute costs by up to 45%**.
- Designed a **secure, multi-tenant data platform** using **AWS IAM, Lake Formation**, and **KMS**, ensuring compliance with **SOC 2 and PCI-DSS** requirements.
- Partnered with **machine learning teams** to deliver streaming feature pipelines supporting real-time inference, anomaly detection, and predictive analytics.
- Built **CI/CD pipelines** using **GitHub Actions, Terraform**, and **Docker**, enabling automated **testing, deployment**, and **rollback** of data infrastructure.
- Defined **SLAs, data quality KPIs**, and **lineage standards** for mission-critical datasets including **transactions, account activity**, and **compliance data**.
- Led **cross-region data replication** and **disaster recovery** strategies using **AWS DMS** and **S3 replication** to meet regulatory availability requirements.

- Enabled **self-service analytics** by integrating **Tableau, Looker, and Power BI** with **Snowflake, BigQuery, and Redshift**.
- Conducted **architecture reviews** and **mentored engineers** on **Spark optimization, distributed systems, and cloud-native best practices**.
- Championed **data observability and reliability** through proactive **monitoring, alerting, and automated runbooks**.
- Collaborated with **product, security, compliance, and clinical stakeholders** to ensure audit readiness and regulatory alignment.
- Played a central role in a company-wide **data modernization initiative**, consolidating fragmented systems into a **governed enterprise lakehouse**.

Accenture
Data Engineer

Chicago, IL
Jan 2018 – Apr 2022

- Designed, developed, and maintained **scalable batch** and streaming **data pipelines** supporting **healthcare, operational, and analytics** use cases across multiple client environments.
- Implemented **Spark-based ETL frameworks** capable of processing large-scale structured and semi-structured datasets with reliability and performance guarantees.
- Led the migration of **on-premise ETL** workflows to **Azure Data Factory** and **Azure Data Lake Storage (ADLS)**, reducing infrastructure costs by **35%** and improving scalability.
- Built **Kafka-based ingestion** pipelines to support near **real-time data availability** for reporting, monitoring, and downstream analytics.
- Designed **dimensional data models, star schemas, and aggregated views** that enabled efficient querying and consistent business metrics across **BI tools**.
- Orchestrated complex, **multi-stage workflows** using **Apache Airflow**, ensuring correct dependency management, retries, and failure handling.
- Collaborated closely with **data scientists** and **analysts** to prepare clean, well-modeled datasets suitable for advanced analytics and machine learning workloads.
- Implemented data masking and encryption using **Azure Key Vault, Azure Purview, and Azure Data Lake Storage access controls**.
- Created a unified data catalog using **Azure Purview** and **Amundsen**, improving **data discoverability** and lineage visibility across teams.
- Implemented **data validation, reconciliation, and monitoring processes** to proactively detect and resolve data quality issues.
- Optimized **Spark jobs** and **SQL queries** through partitioning strategies, indexing, and query tuning to improve performance and reduce processing costs.
- Supported the **migration of legacy, on-premise data systems** to **cloud-based data lakes and warehouses**.
- Worked directly with product owners and business stakeholders to translate analytical requirements into scalable technical solutions.
- Contributed to the establishment of data governance practices including documentation, lineage tracking, and access controls.
- Participated in **on-call rotations, diagnosing** and resolving production **data pipeline** issues under **SLA constraints**.
- Produced comprehensive technical documentation covering data architectures, pipeline logic, and operational procedures.
- Collaborated with engineering teams to **standardize data modeling** and **pipeline design patterns**.
- Continued to deepen expertise in **distributed systems, cloud data platforms, and large-scale data processing**.

Optum
Junior Data Engineer

Eden Prairie, MN
Jul 2015 – Dec 2017

- Developed and maintained **batch ETL pipelines** integrating data from transactional, operational, and third-party systems into centralized analytics stores.
- Built and optimized **SQL-based data transformations** to support reporting, analytics, and operational decision-making.
- Assisted in the design and maintenance of **enterprise data warehouse schemas**, including **star** and **snowflake models**.
- Tuned long-running **ETL jobs** to improve throughput, reduce failures, and meet reporting deadlines.
- Collaborated with senior engineers to debug complex data issues and optimize pipeline reliability.
- Ensured data accuracy, consistency, and completeness through validation and reconciliation processes.

- Supported **analytics** and reporting teams by delivering curated datasets tailored to business needs.
- Participated in operational support and maintenance of data infrastructure.
- Contributed to documentation and internal knowledge-sharing initiatives.
- Built a strong technical foundation in **data engineering, SQL, and enterprise data systems.**

EDUCATION

University of Southern California

Los Angeles, CA

Bachelor of Computer Science

2010 - 2014